



CSS 3

SESSION 4

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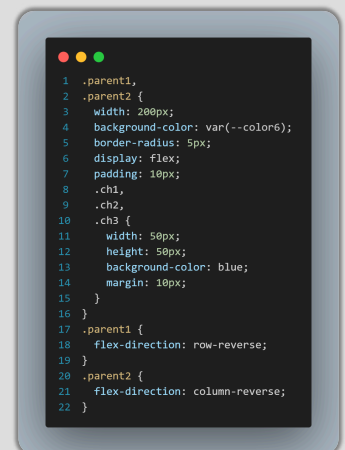
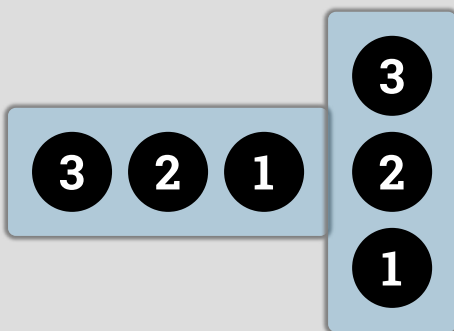
Flexbox

display - flex-direction - flex-wrap

Parent :

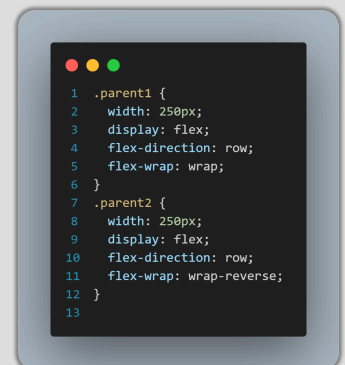
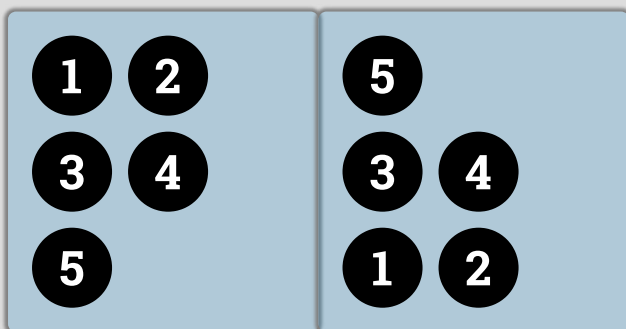
display: flex (When you apply it to a container, all its direct children become **flex items**, which you can easily control)

- width : auto
- height : stretch
- **flex-direction**: [row "default" - row-reverse - column - column-reverse]



flex-wrap : is used with display: flex to control whether flex items should stay in one line or wrap onto multiple lines when they don't fit in the container.

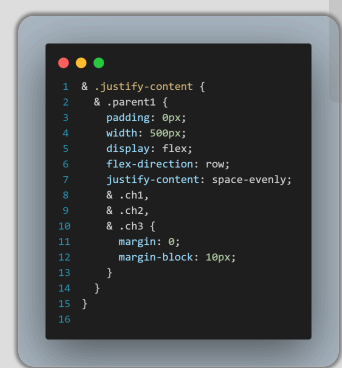
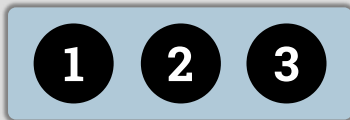
- flex-wrap value [nowrap "default" - wrap - wrap-reverse]



justify-content - align-item - align-content

justify-content is a Flexbox property that controls the horizontal alignment of flex items inside a flex container (if the direction is row).

- justify-content : [start "default" - end - center - space-between - space-around - space-evenly]



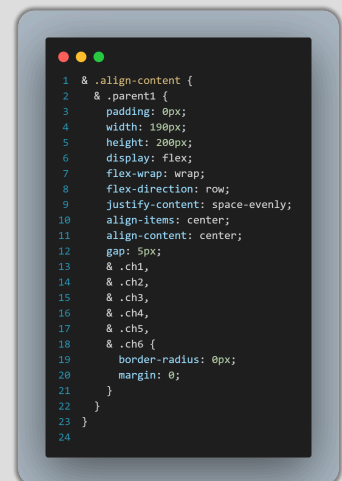
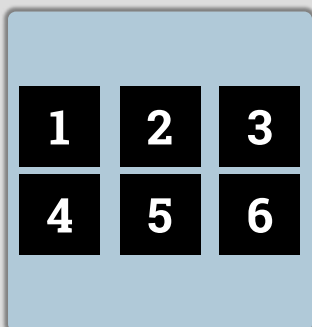
align-items It controls how flex items are aligned vertically

- align-items: [stretch - start - end - center]



align-content : controls the spacing and alignment between multiple flex lines (rows or columns) along the cross axis.

- align-content: [stretch - start - end - center - space-between - space-around - space-evenly]



Note :

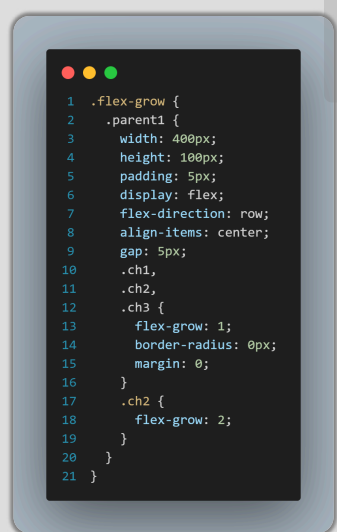
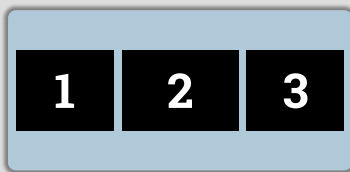
- align-items : aligns items inside one flex line (works with single row).
- align-content : aligns multiple flex lines (only works if flex-wrap: wrap creates more than one row).

children :

grow - shrink - order

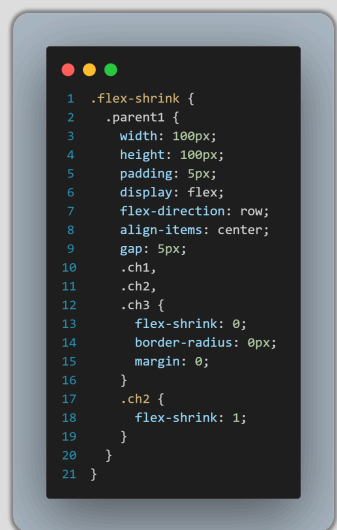
flex-grow : property in CSS defines how much a flex item should grow relative to the rest of the items inside the same flex container when there is extra space available.

- 0 (default) : The item will not grow; it keeps its base size.
- Positive number (1, 2, 3, etc.) : The item will grow to fill the remaining space proportionally compared to other items.



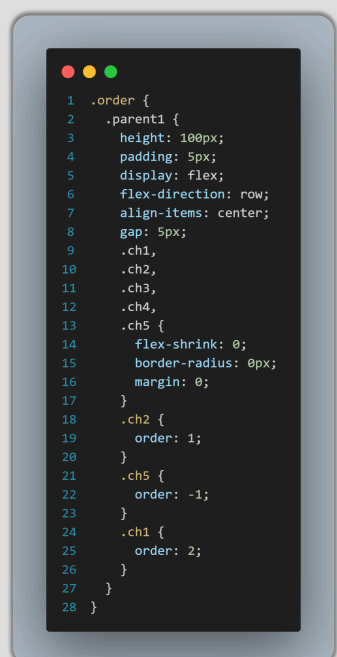
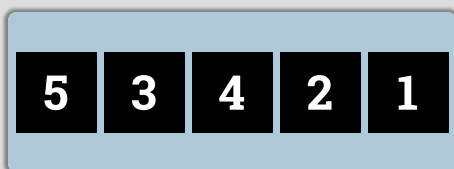
flex-shrink : property in CSS defines how much a flex item will shrink relative to the rest of the flex items when there isn't enough space in the flex container.

- 0 : The item will not shrink even if there's not enough space.
- Positive number (1, 2, 3, etc.) : The item will shrink relative to other items when space is tight. The higher the number, the more it shrinks.



order : property controls the visual order of flex items inside a flex container — without changing the actual order of the HTML elements in the code.

- Default value: 0
- Type: integer (can be positive, negative, or zero)



basis - flex - inline-flex



flex-basis :the initial main size (width or height) of a flex item before the remaining space is distributed using flex-grow or flex-shrink.

- as a width with **flex-direction : row**
- as a height with **flex-direction : column**
- value : auto - length - percentage - content
- flex-basis is super useful for **responsive design** – because it lets you control the starting size of flex items relative to the container, not fixed pixels only.



Note : flex-basis : auto with flex-grow : 1

- Start at natural (auto) size → then grow equally to fill extra space.

```
1 @media (min-width :780px) {
2   .parent {
3     width: 90%;
4     height: 250px;
5     border: 1px solid;
6     background-color: #ddd;
7     margin: auto;
8     display: flex;
9     padding: 10px;
10    gap: 10px;
11    .child {
12      flex-basis: auto;
13      border: 1px solid;
14      flex-grow: 1;
15    }
16    .one {
17      background-color: red;
18      color: #ff0;
19      font-size: 50px;
20    }
21    .two {
22      background-color: green;
23    }
24    .three {
25      background-color: blue;
26    }
27  }
28 }
```

Position

Static - Fixed - Absolute - Relative - Sticky

top - bottom - left - right

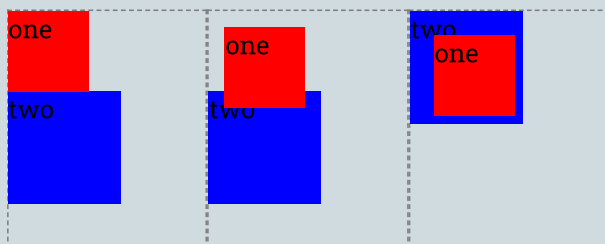
Position : Static is a default value using for reset

position: relative property in CSS means the element is positioned relative to its normal position in the page and you can use negative value.

position: absolute property in CSS means the element is completely removed from the normal document flow and is positioned relative to the nearest ancestor that has a position value other than static (for example relative, absolute, fixed, or sticky).

position: fixed property in CSS means the element is positioned relative to the browser window (viewport) – not to any parent element – and it does not move when you scroll the page.

position: sticky property in CSS means the element acts like relative at first, but becomes fixed once you scroll past a certain point – usually defined by top, bottom, left, or right.



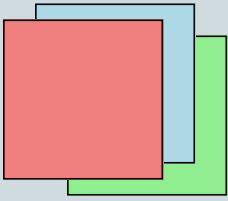
position: static; top: 10px; left: 10px;directional properties not work
position: relative; top: 10px; left: 10px;
position: absolute; top: 15px; left: 15px; give the parent position: relative to move inside it

Note: when you give div one margin-top 20px that make div two move down 20px

Z-Index



The z-index property in CSS controls the stacking order of elements that overlap each other. In simple terms — it decides which element appears on top and which one appears below.



```
1 .one,.two,.three {  
2   width: 100px;  
3   height: 100px;  
4   border: 1px solid;  
5   position: absolute;  
6 }  
7 .one {  
8   background-color: lightblue;  
9   top: 30px;  
10  left: 30px;  
11  z-index: 10;  
12 }  
13 .two {  
14  background-color: lightcoral;  
15  top: 40px;  
16  left: 20px;  
17  z-index: 11;  
18 }  
19 .three {  
20  background-color: lightgreen;  
21  top: 50px;  
22  left: 50px;  
23  z-index: 9;  
24 }
```

Note : The z-index property won't work unless the element has a position value other than static.